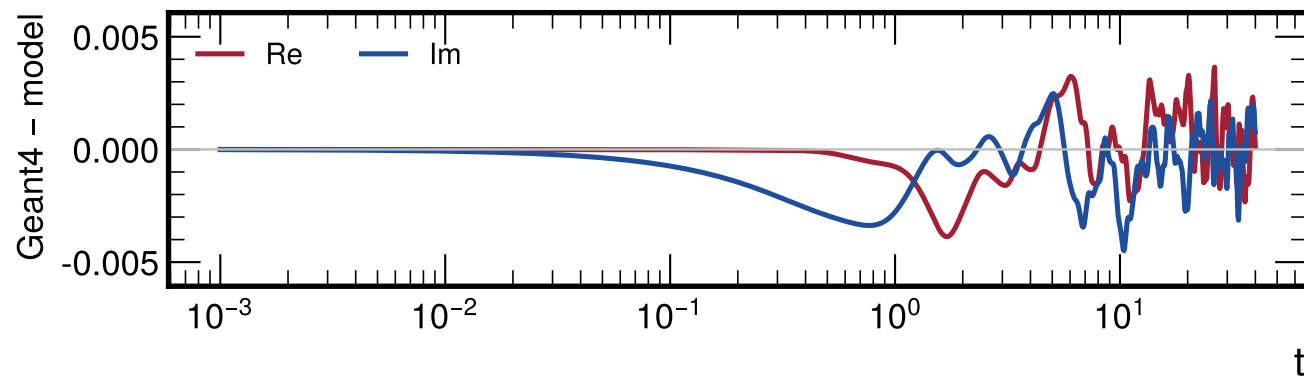
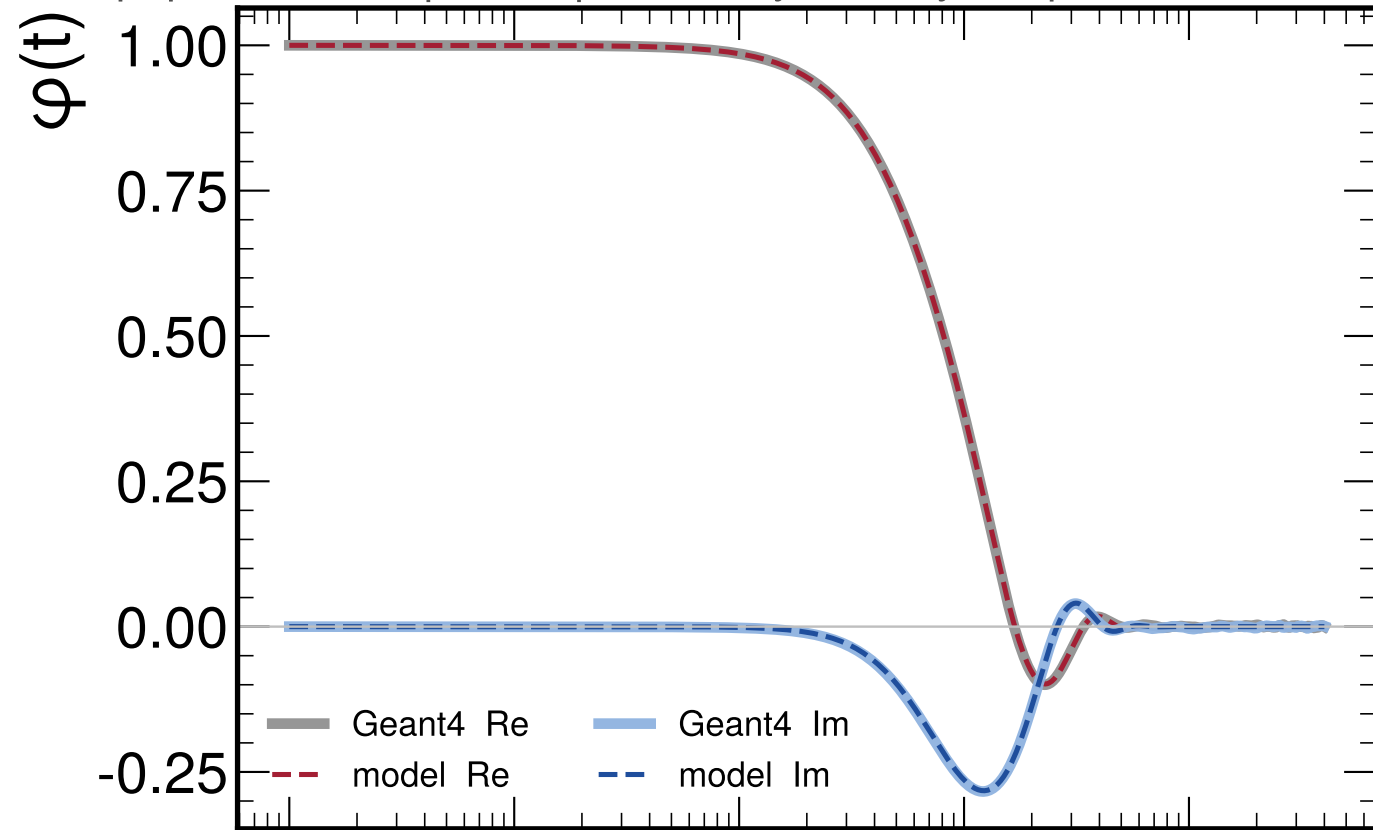
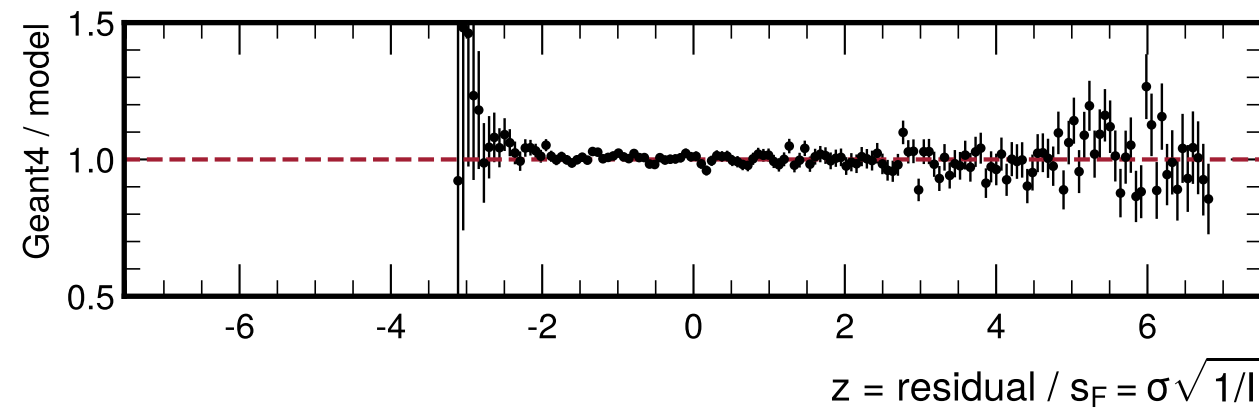
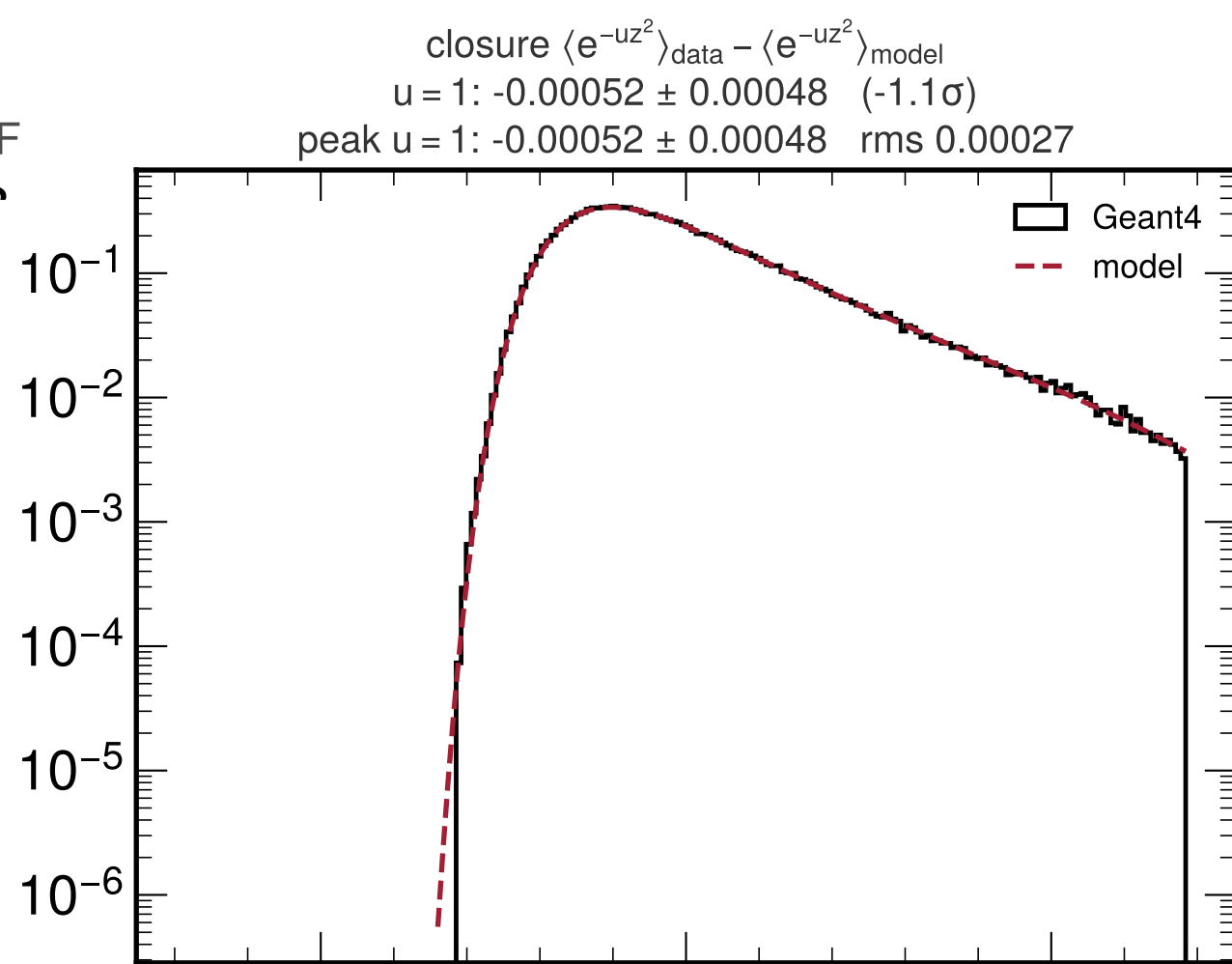


q/p at r = 107 cm — ionization straggling
 $p, p_T = 3 \text{ GeV}, \eta = 0.3, \phi = 0.7 \cdot \text{layered toy, 14 planes} \cdot \text{radiation OFF}$



density



closure $\langle e^{-uz^2} \rangle_{\text{data}} - \langle e^{-uz^2} \rangle_{\text{model}}$
 $u = 1: -0.00052 \pm 0.00048 \quad (-1.1\sigma)$
peak $u = 1: -0.00052 \pm 0.00048 \quad \text{rms } 0.00027$

$z = \text{residual} / s_F = \sigma \sqrt{1/l}$